

Solution to Exercise 4.12 (Page 81)

$$\text{Problem 1)} \begin{cases} \min_{x \in \mathbb{R}} & x^4 \\ \text{s.t.} & x^2 \geq 1 \end{cases}$$

- $f^* = 1, x^* = \pm 1, \mu^* = 2$
- $h(x) = 1 - x^2 \leq 0$ strictly concave $\Rightarrow$ non-convex problem
- $L(x, \mu) = x^4 + \mu(1 - x^2)$
- Dual function $\theta(\mu) = \inf_x L(x, \mu)$:

$$\nabla_x L(x, \mu) = 2x(2x^2 - \mu) = 0 \text{ leads to}$$

$$\mu \in \mathbb{R} : \quad x_1 = 0 \quad \text{with} \quad L(x_1, \mu) = \mu$$

$$\mu > 0 : \quad x_{2,3} = \pm \sqrt{\frac{\mu}{2}} \quad L(x_{2,3}, \mu) = \mu - \frac{\mu^2}{4}$$

$$\Rightarrow \theta(\mu) = \begin{cases} \mu - \frac{\mu^2}{4} & \mu \geq 0 \\ \mu & \text{else} \end{cases}$$

- $\theta^* = \max_{\mu \geq 0} \theta(\mu)$: $\theta'(\mu) = 0$ gives $\mu = 2$ with $\theta^* = \theta(2) = 1$
- Strong duality ($f^* = \theta^*$) despite $h(x) = 1 - x^2 \neq \text{convex}$!

$$\text{Problem 2)} \begin{cases} \min_{x \in \mathbb{R}} & x^2 \\ \text{s.t.} & x^4 \geq 1 \end{cases}$$

- $f^* = 1, x^* = \pm 1, \mu^* = \frac{1}{2}$, again non-convex problem
- $L(x, \mu) = x^2 + \mu(1 - x^4)$
- $\theta(\mu) = \inf_x L(x, \mu) = \begin{cases} \mu & \text{if } \mu \leq 0 \\ -\infty & \text{else} \end{cases}$
- $\theta^* = \max_{\mu \geq 0} \theta(\mu) = 0$
- Duality gap due to $\theta^* < f^*$

Conclusions:

- Convexity and Slater's constraint qualification are only sufficient for strong duality (see Problem 1 and Example 4.9)
- Strong duality is essential for equivalence of primal and dual problem